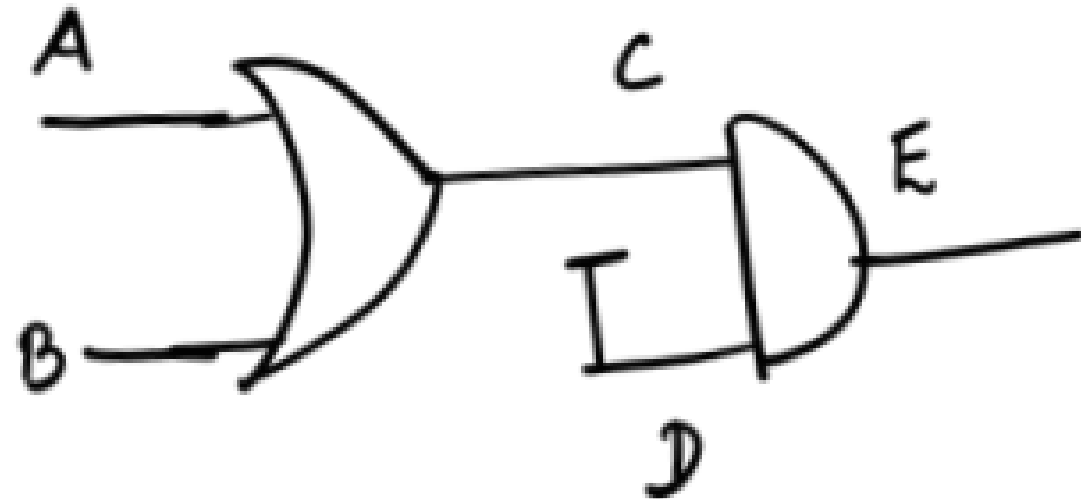
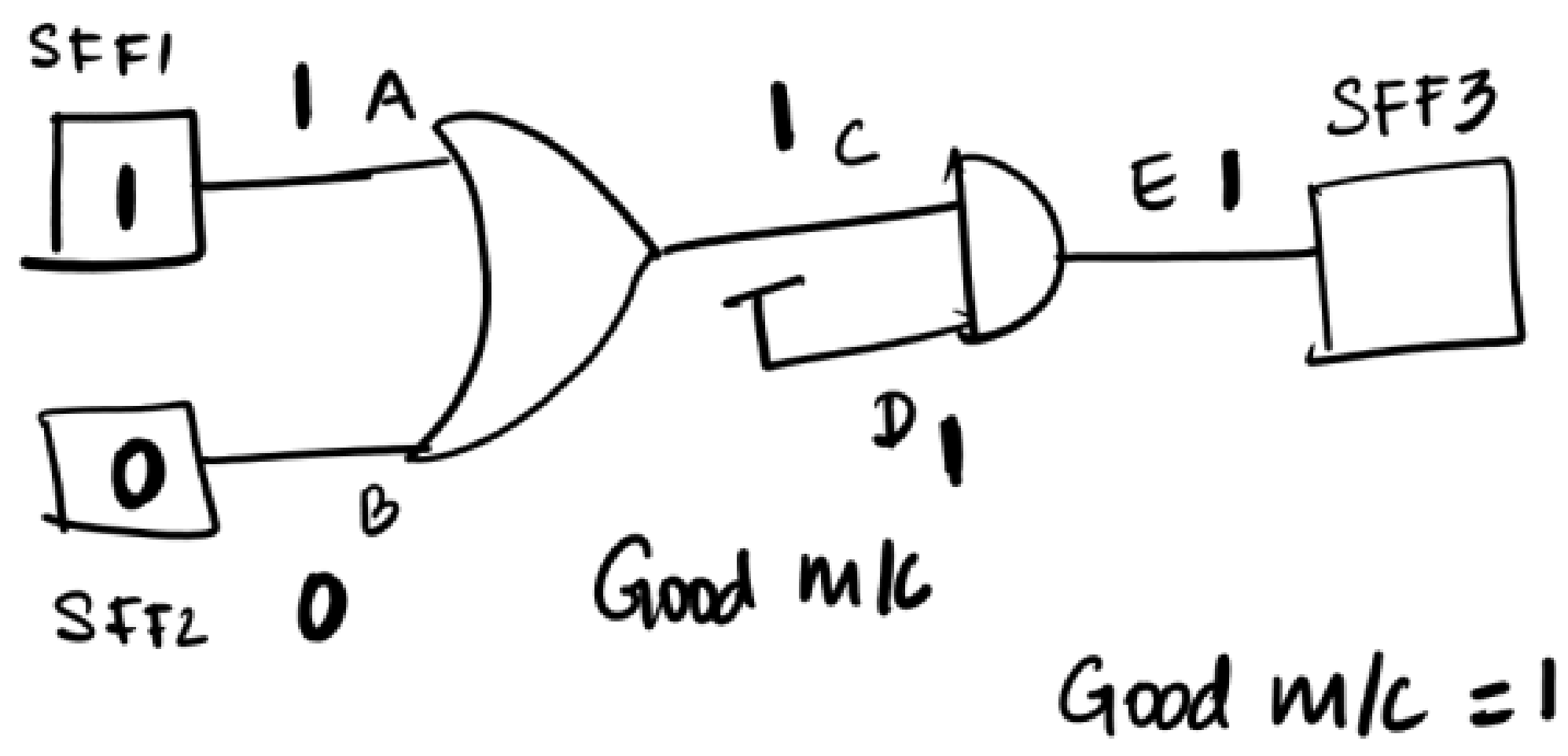


Q:  (a) For this circuit, how many patterns will be generated?

(b) What will be the test coverage and fault coverage?

A	B	ASAO	ASAI	BSAO	BSAI	CSAO	CSAI	DSAO	DSAI	ESAO	ESAI
0	0	X	✓	X	✓	X	✓	X	X	X	✓
0	1	X	X	✓	X	✓	X	✓	X	✓	X
1	0	✓	X	X	X	✓	X	✓	X	✓	X
1	1	X	X	X	X	✓	X	✓	X		

(a) $\{10, 00, 01\} \Rightarrow 3$ patterns will be generated



ASAO:

Assume a fault: ASAO

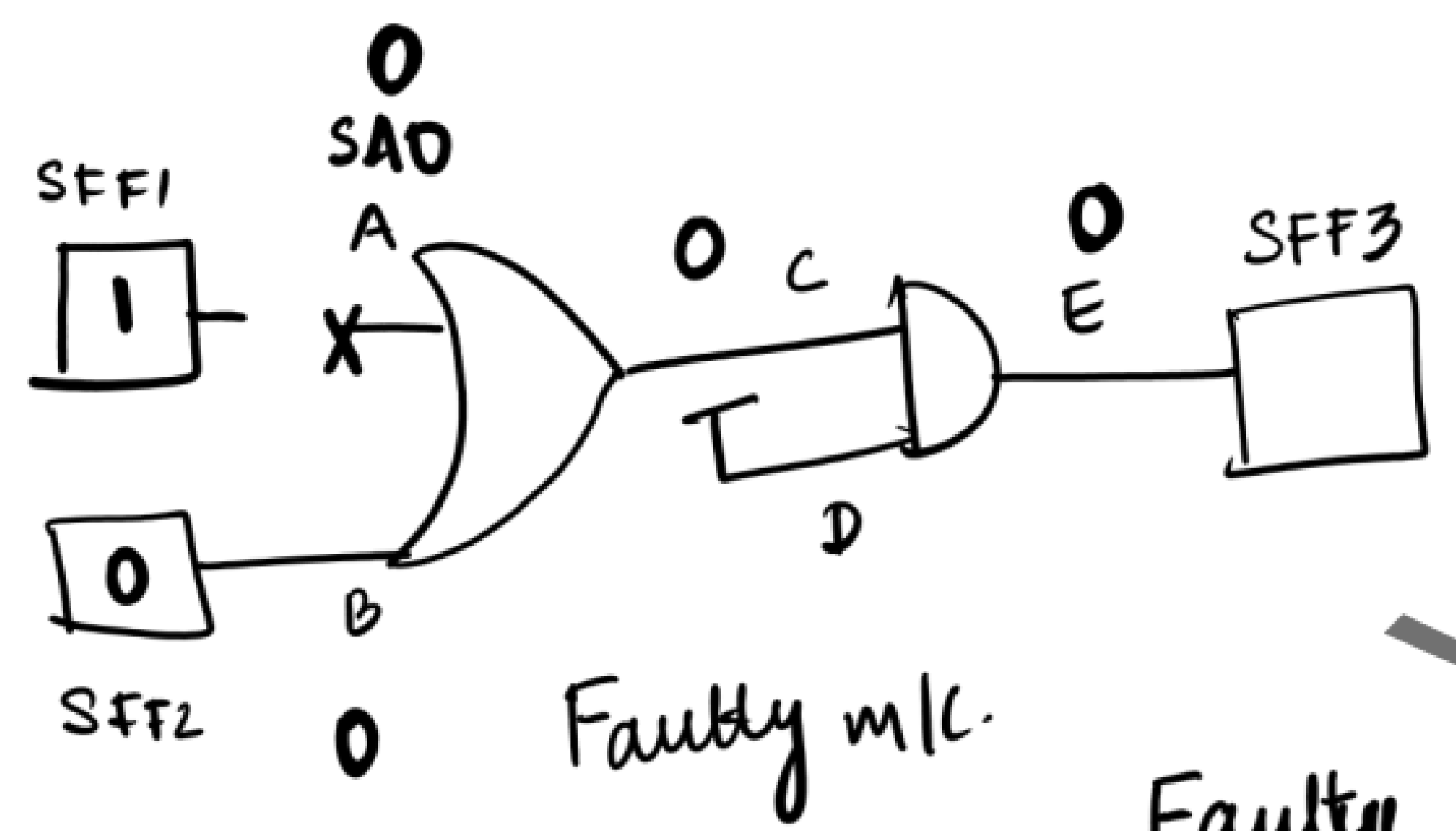
Identify opp value: $A = 1$

Propagate the value at the fault site to an obs pt.

$B = 0$

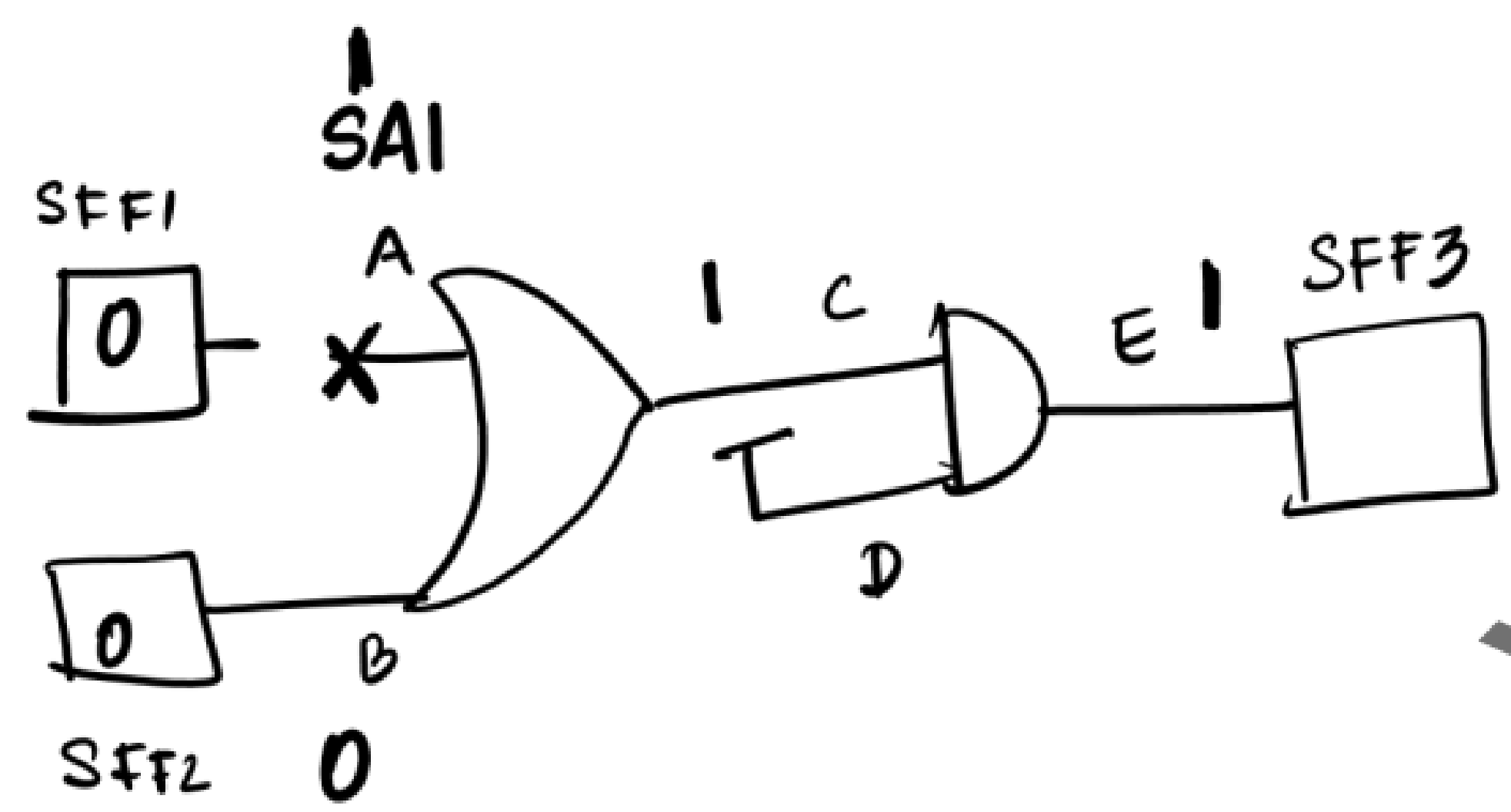
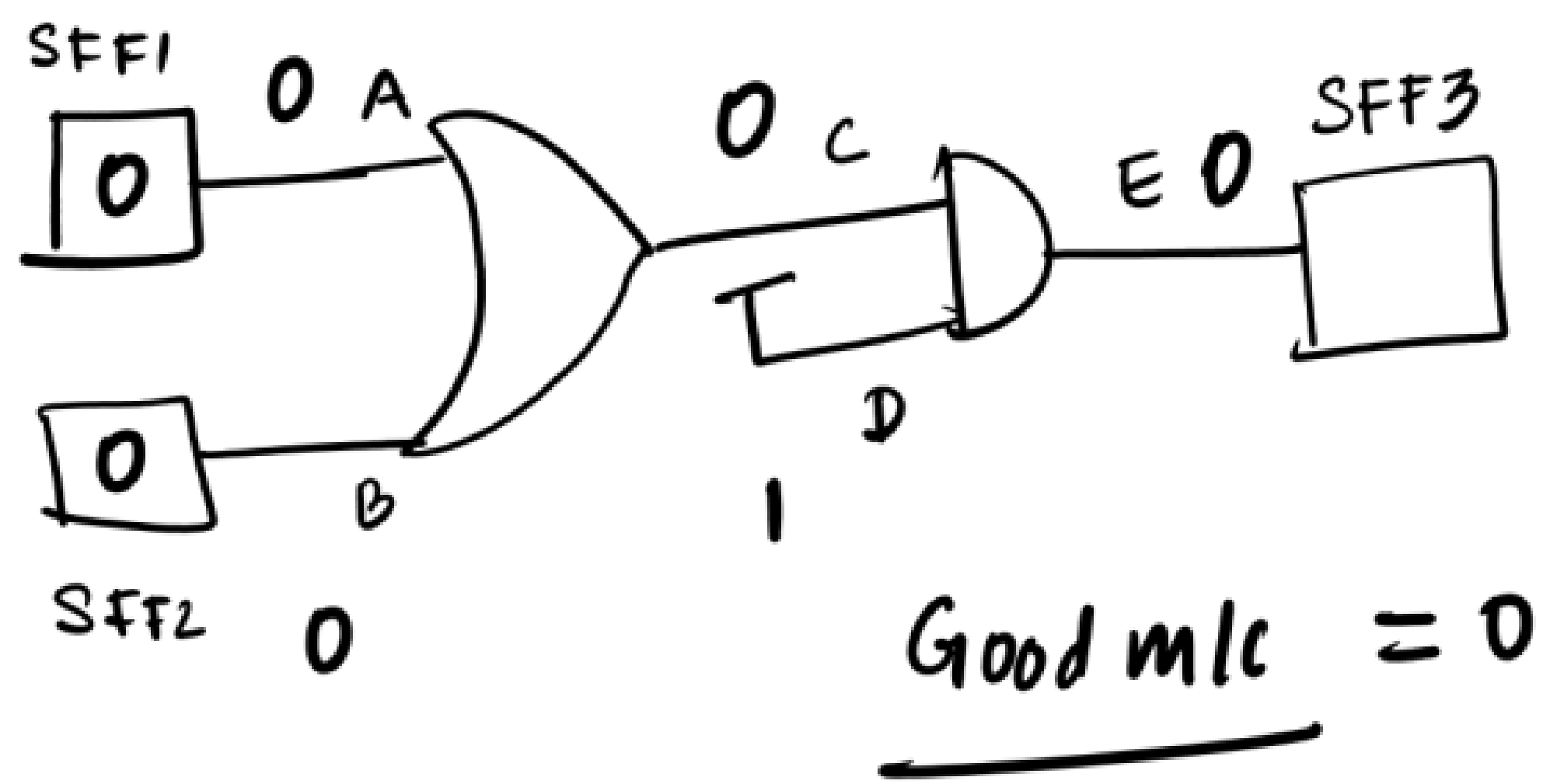
pattern $\Rightarrow A = 1$

$B = 0$



Faulty m/c = 0

Good m/c value $\neq$ Faulty m/c value



ASAI :

Assume a fault: ASAI
 Identify opp value: $A = 0$

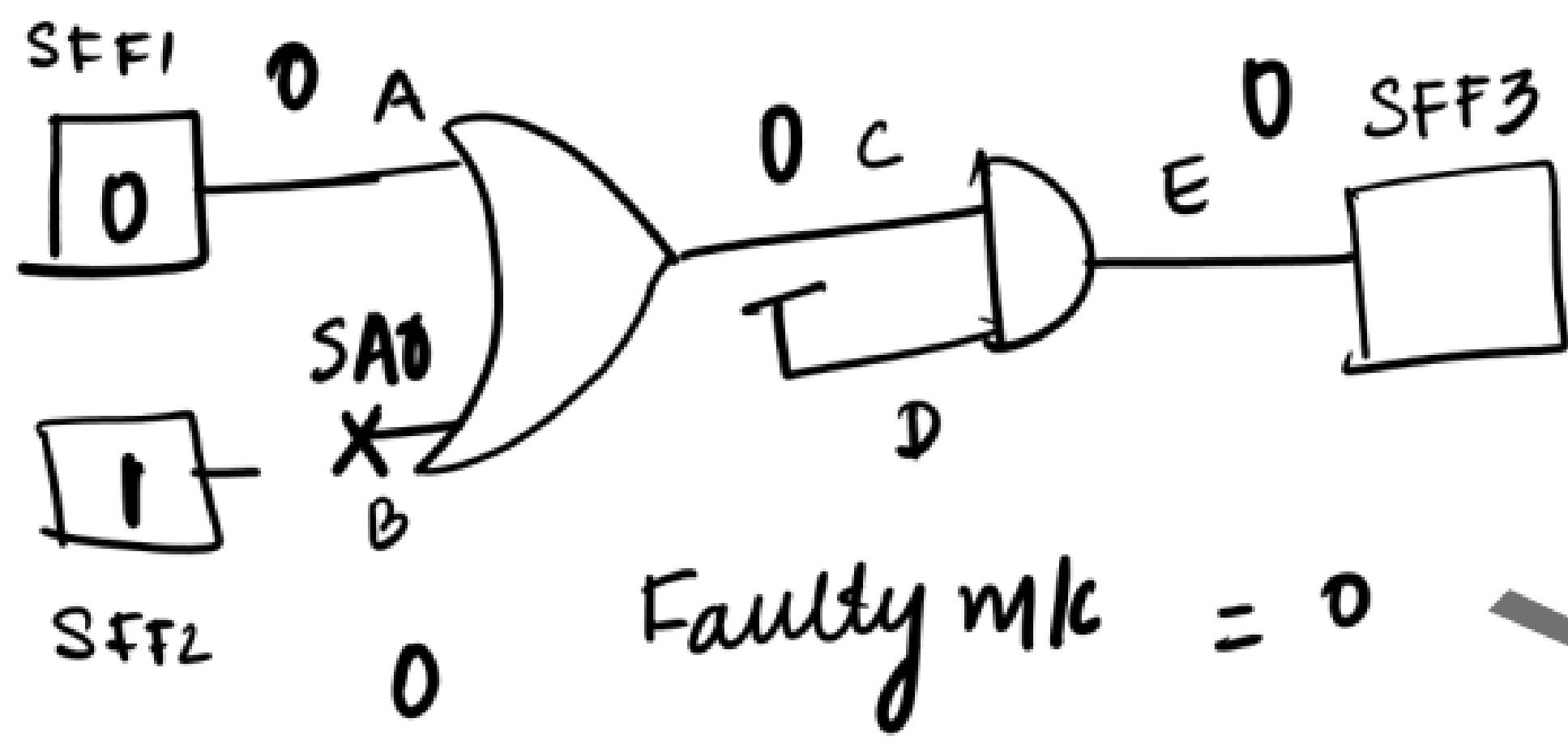
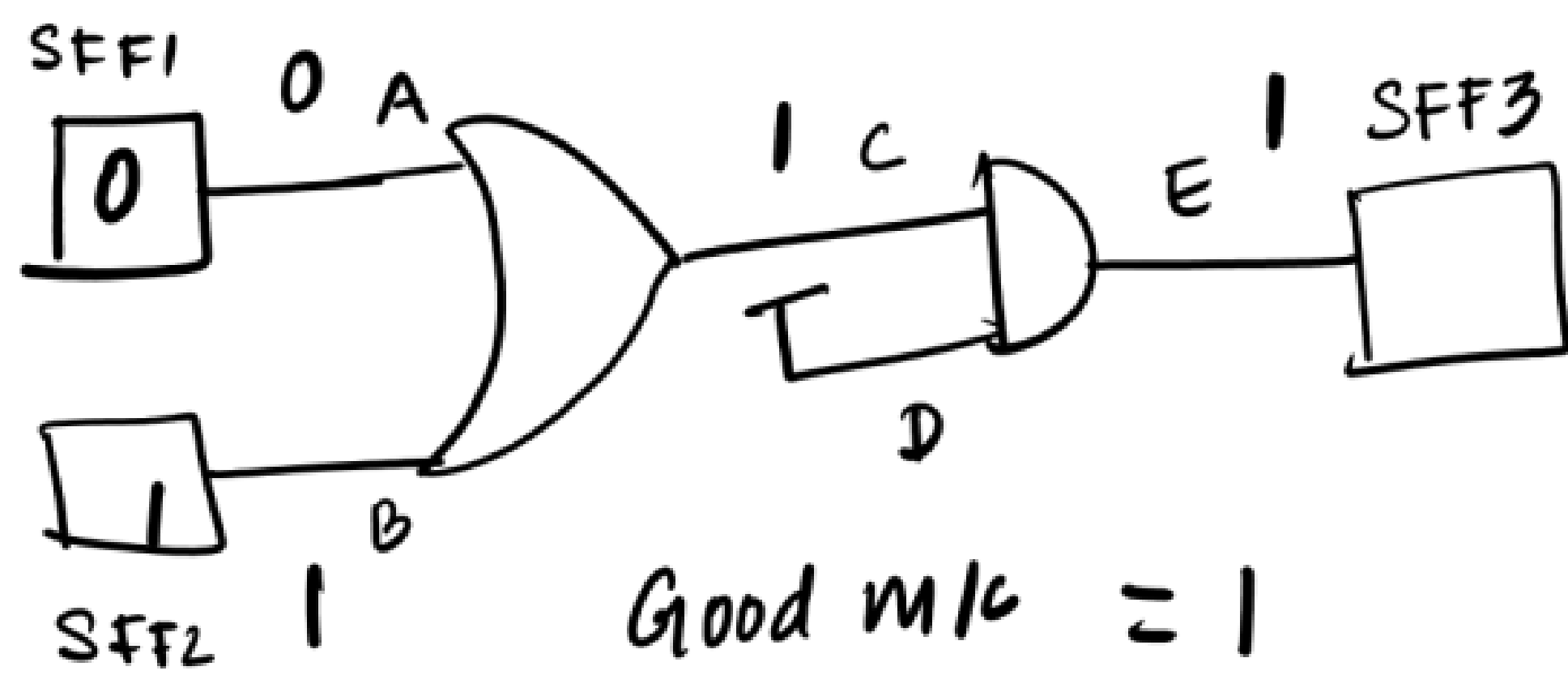
propagate the value at fault site
 to an obs pt
 $B = 0$

$A = 0$

$B = 0$

(B)		(A)	
i ₁	i ₂		y
0	0		0
0	1		1
1	0		1
1	1		1

$Gmc \neq Fmc$



$Gm/c \neq Fm/c$

BSAD :-

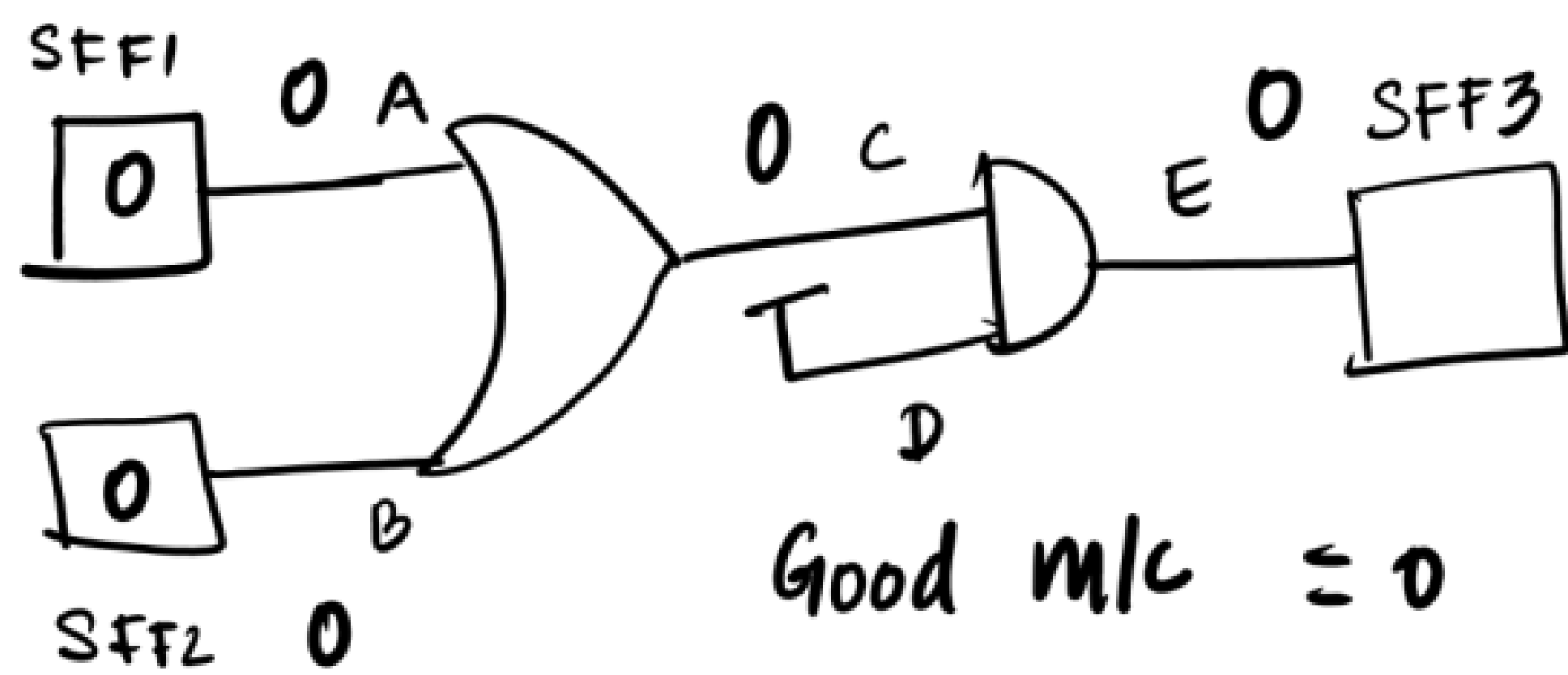
Assume a fault: BSAD

identify opp value: $B = 1$

propagate the value at the fault site to an obs pt

$A = 0$

pattern: $A = 0$
 $B = 1$



BSA1 :

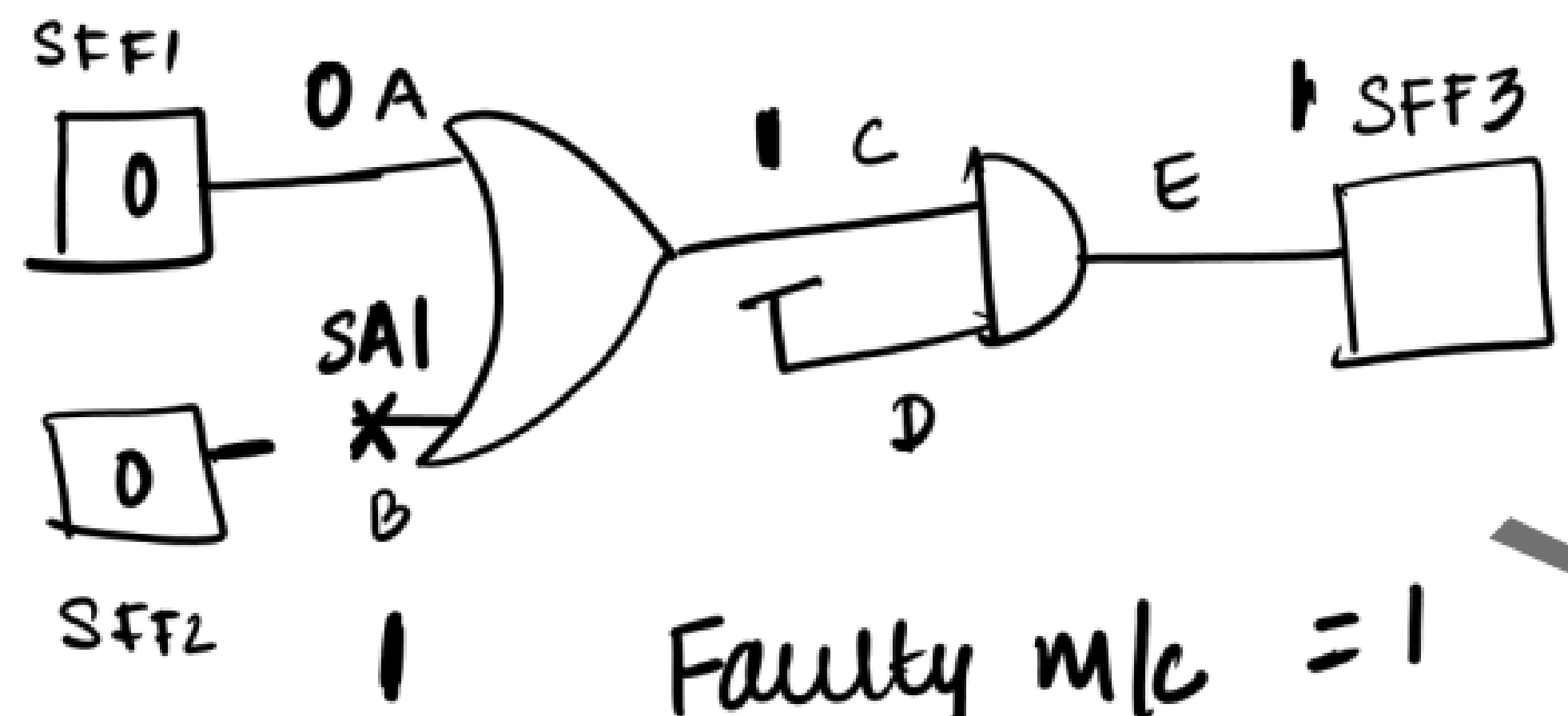
Assume a fault: BSA1
 identify opp value: $B = 0$

prop the value at the fault
 site to an obs pt

$A = 0$

$A = 0$

$B = 0$



$Gm/c \neq Fm/c$

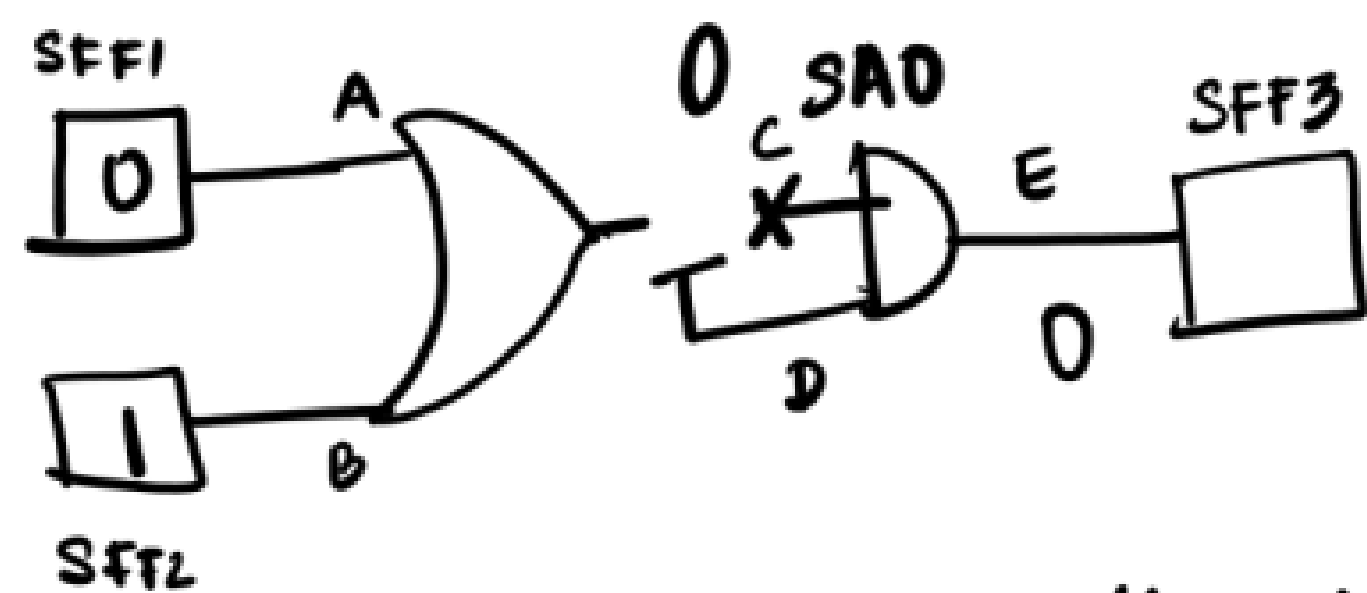
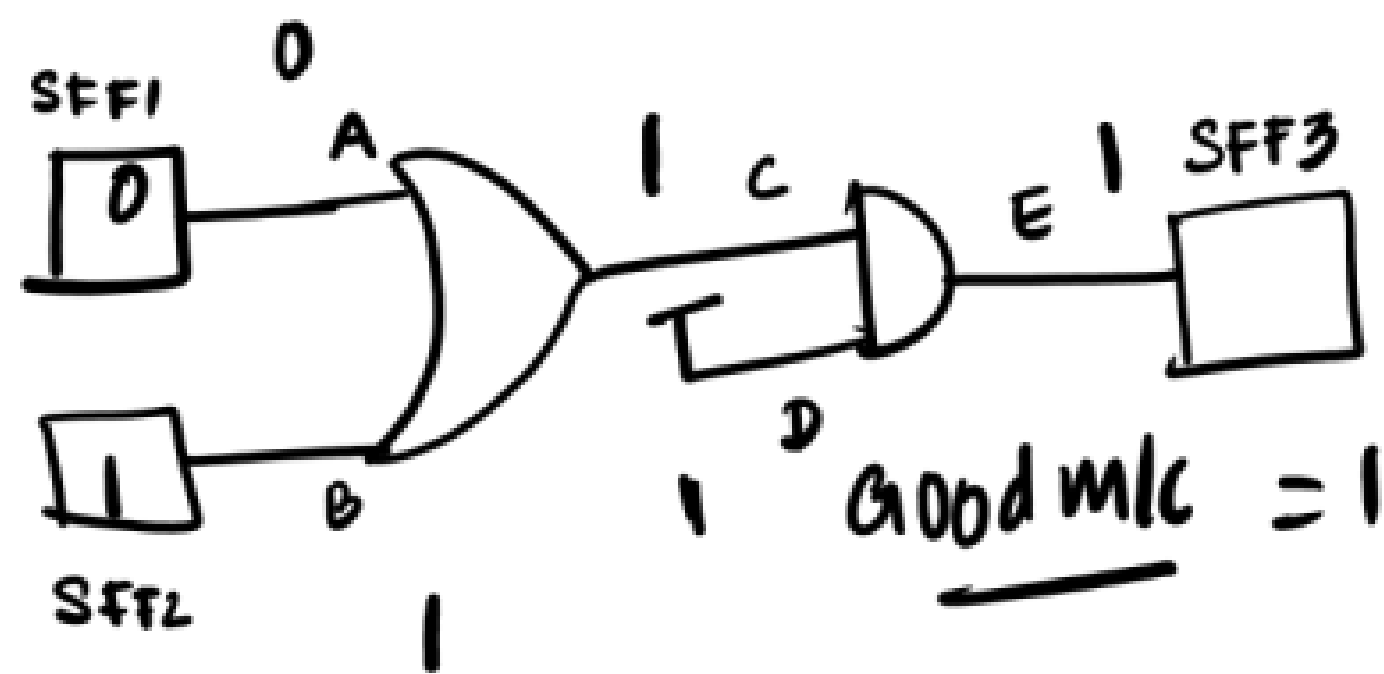
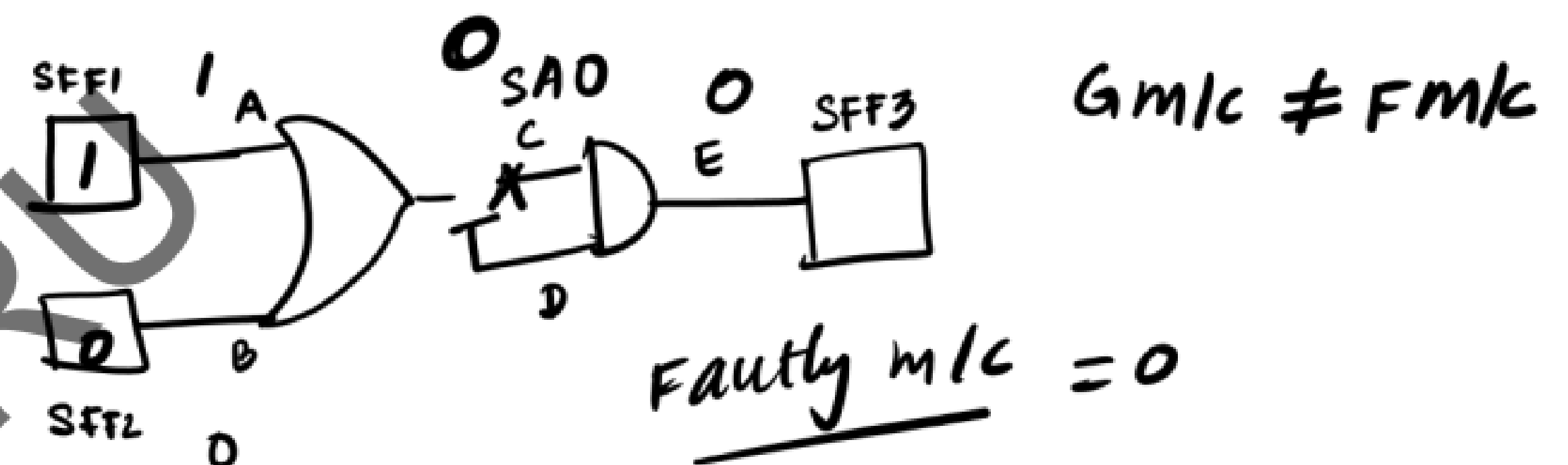
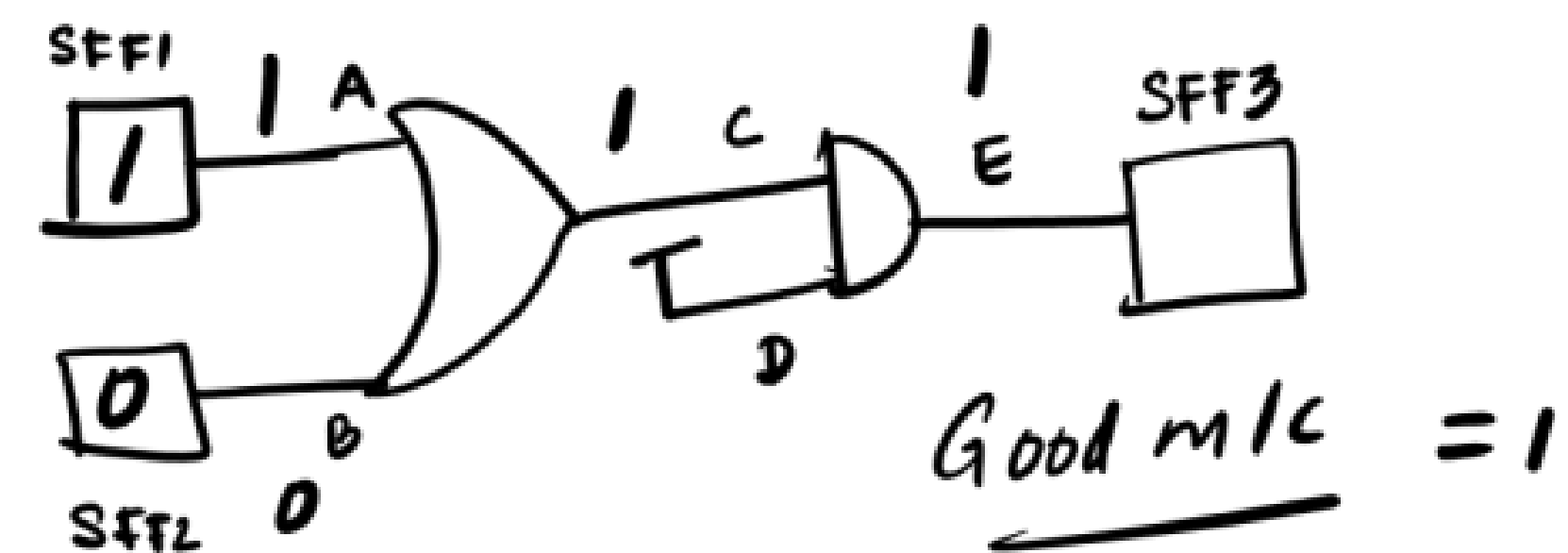
CSAO:

Assume a fault: CSAO

Identify opp value: $c=1$

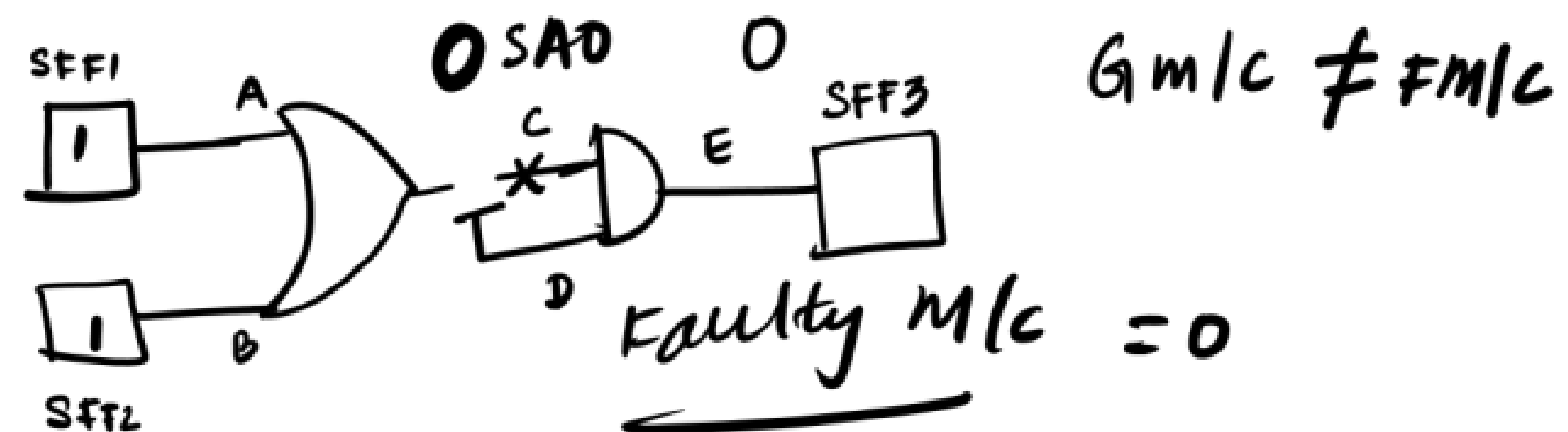
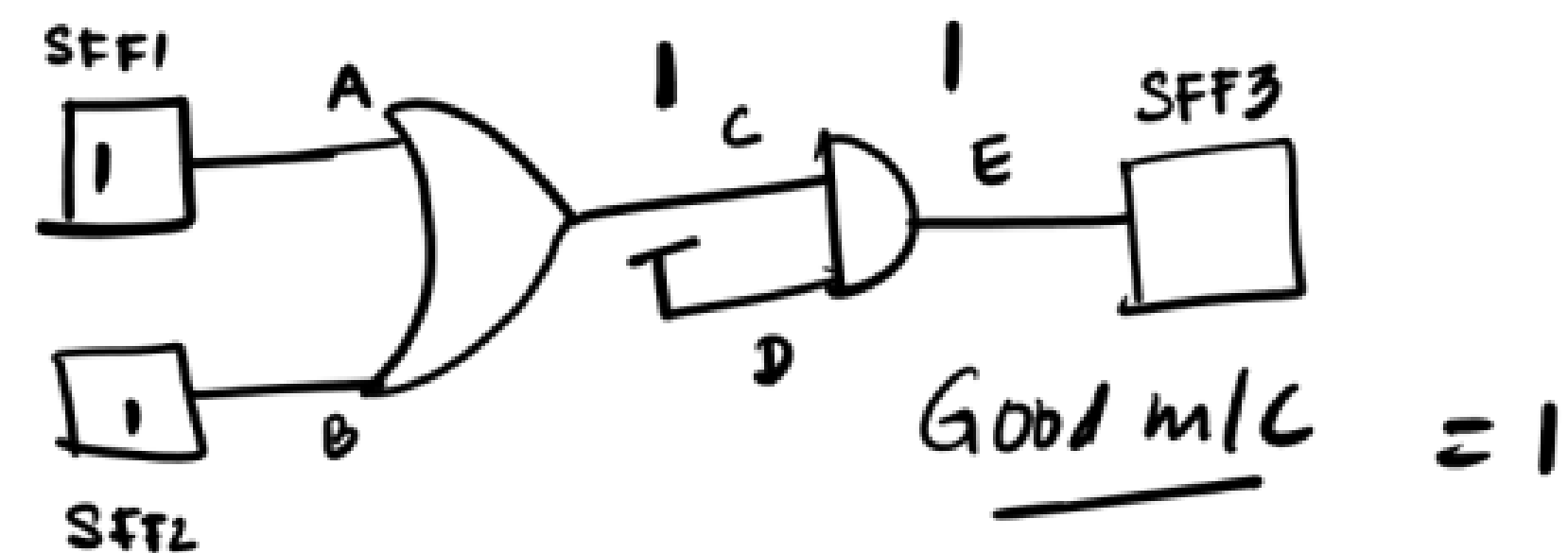
write i/p seq to get the opp value

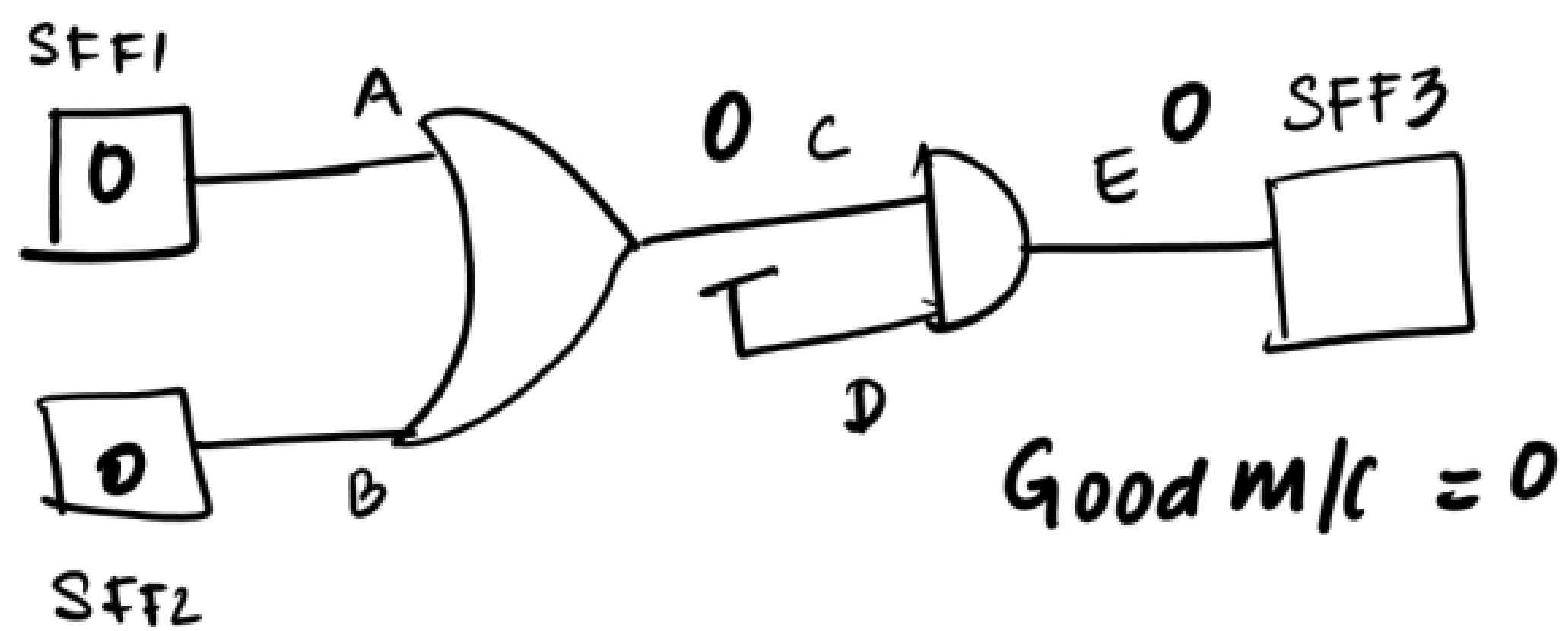
$A, B = (0, 1) \text{ (or) } (1, 0) \text{ (or) } (1, 1)$



Faulty m/c = 0

Gm/c ≠ Fm/c



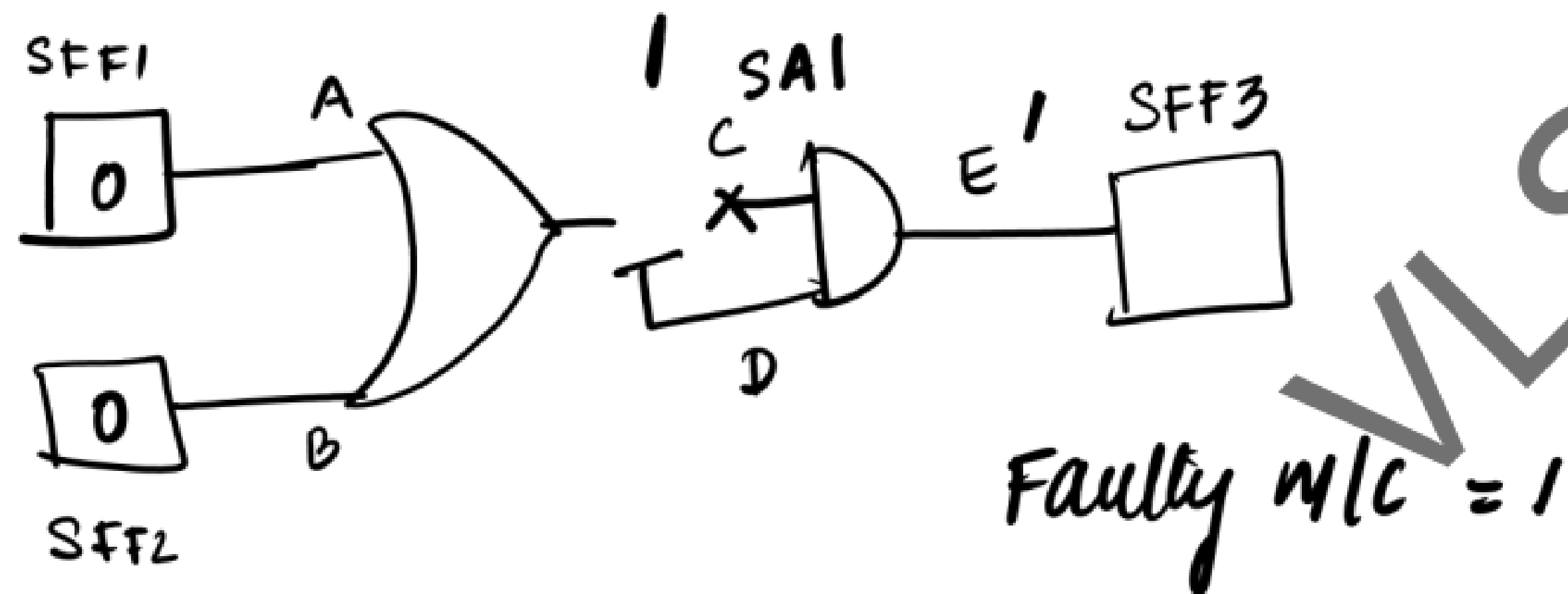


CSA1 :-

Assume a fault: CSA1

Identify opp value: $C = 0$

write i/p seq to get one } $A = 0$
opp value } $B = 0$



$Gm \neq Fm/c$

DSA0:

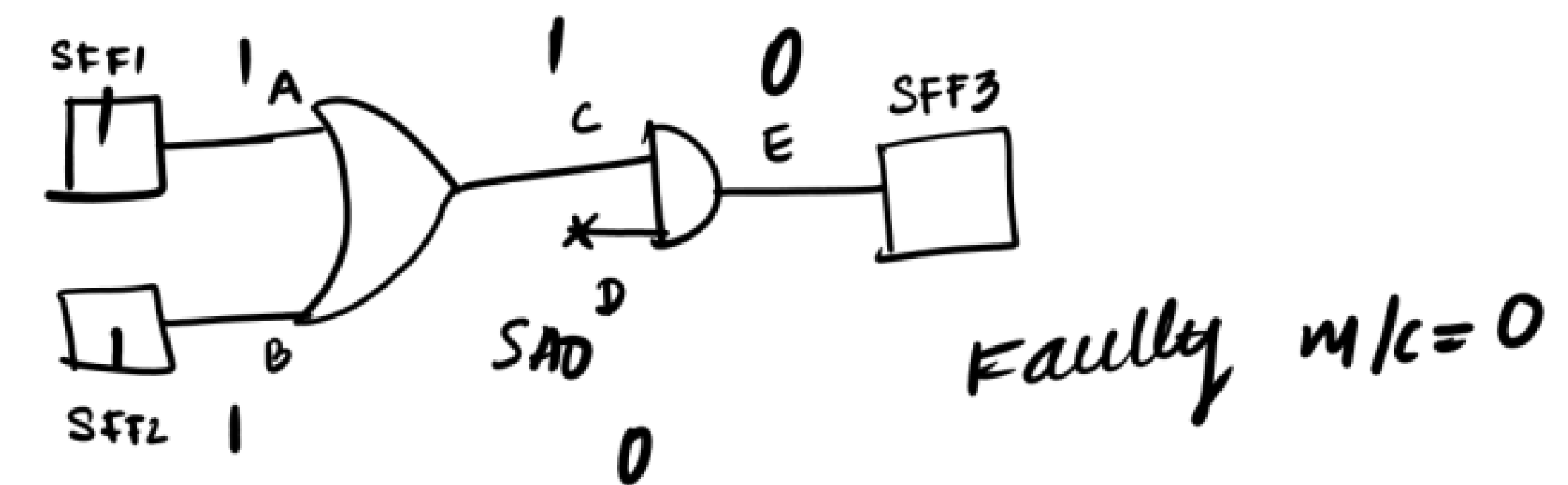
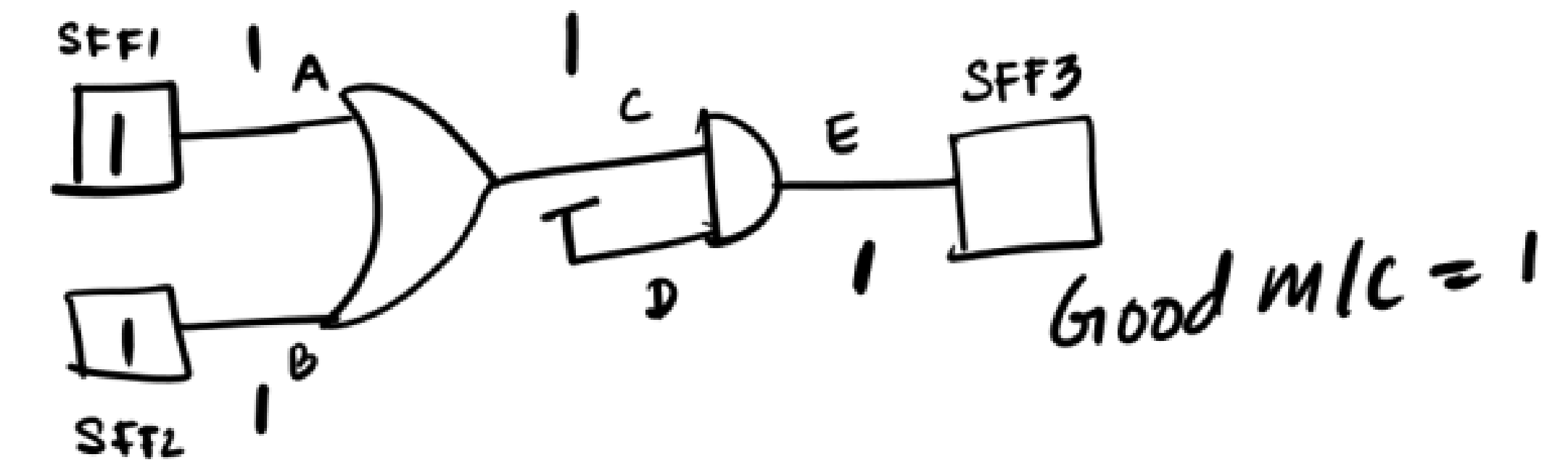
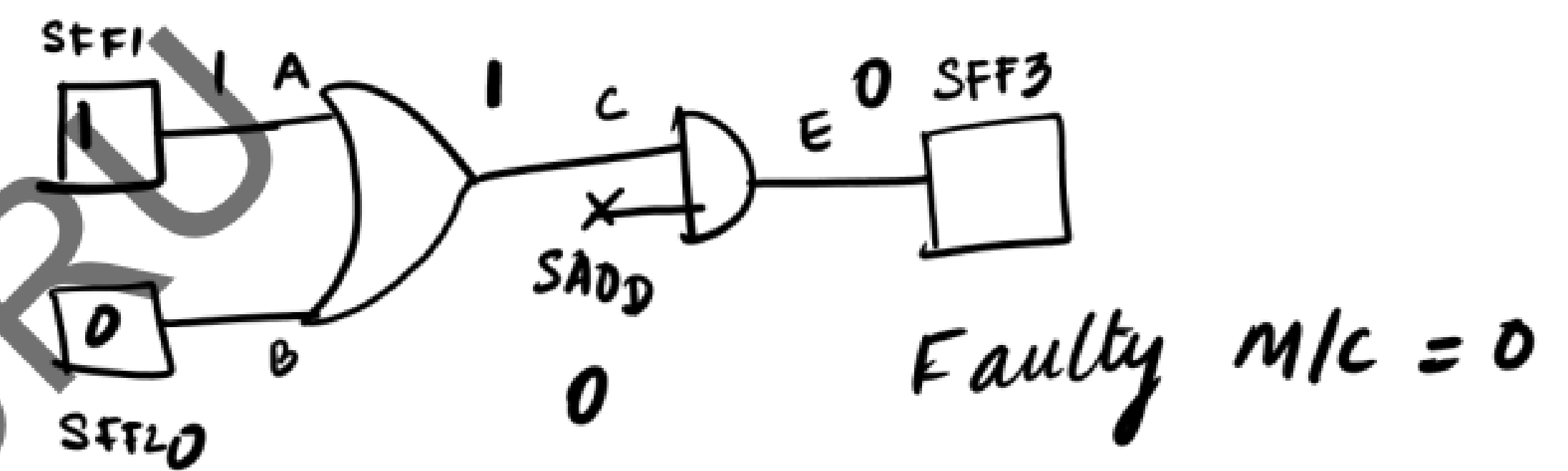
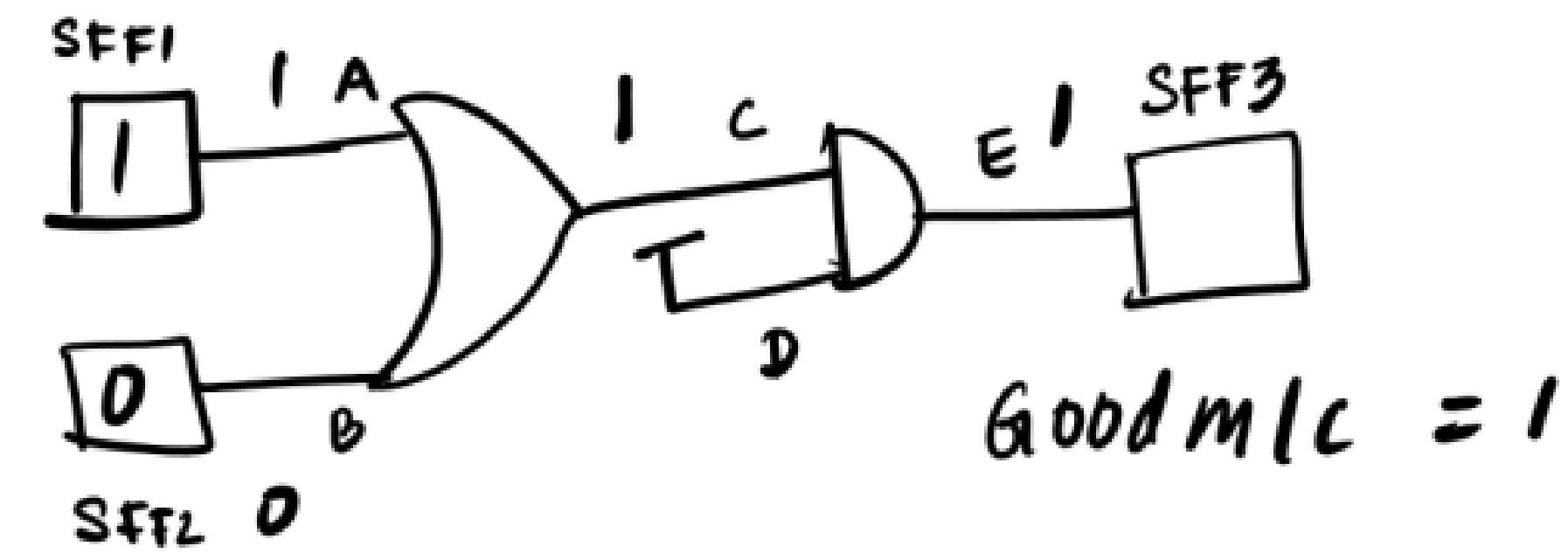
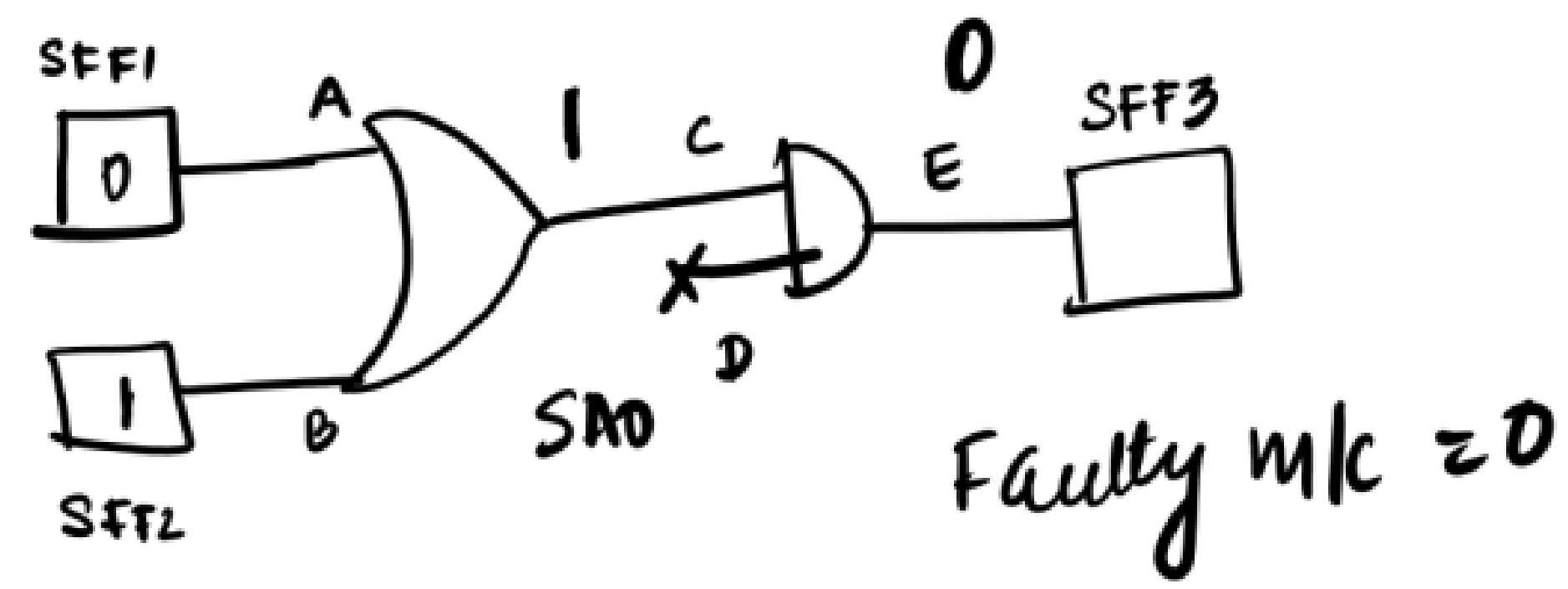
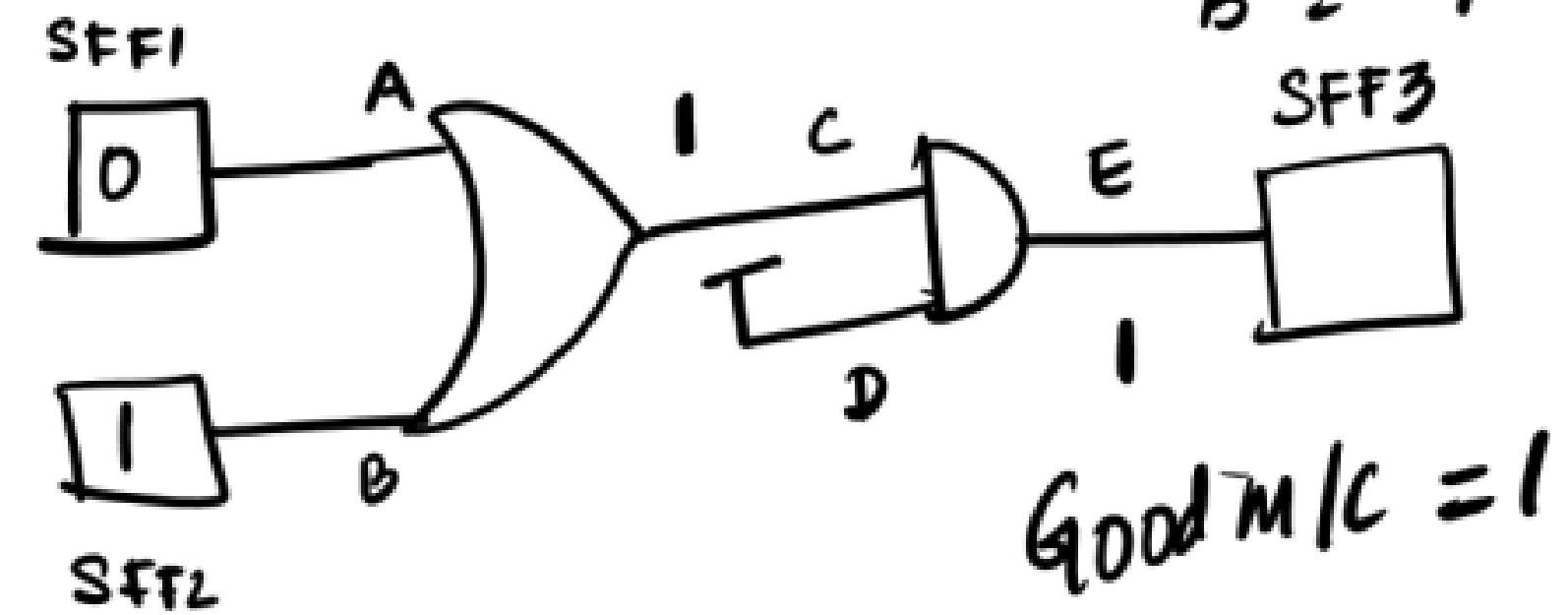
Assume a fault: DSA0

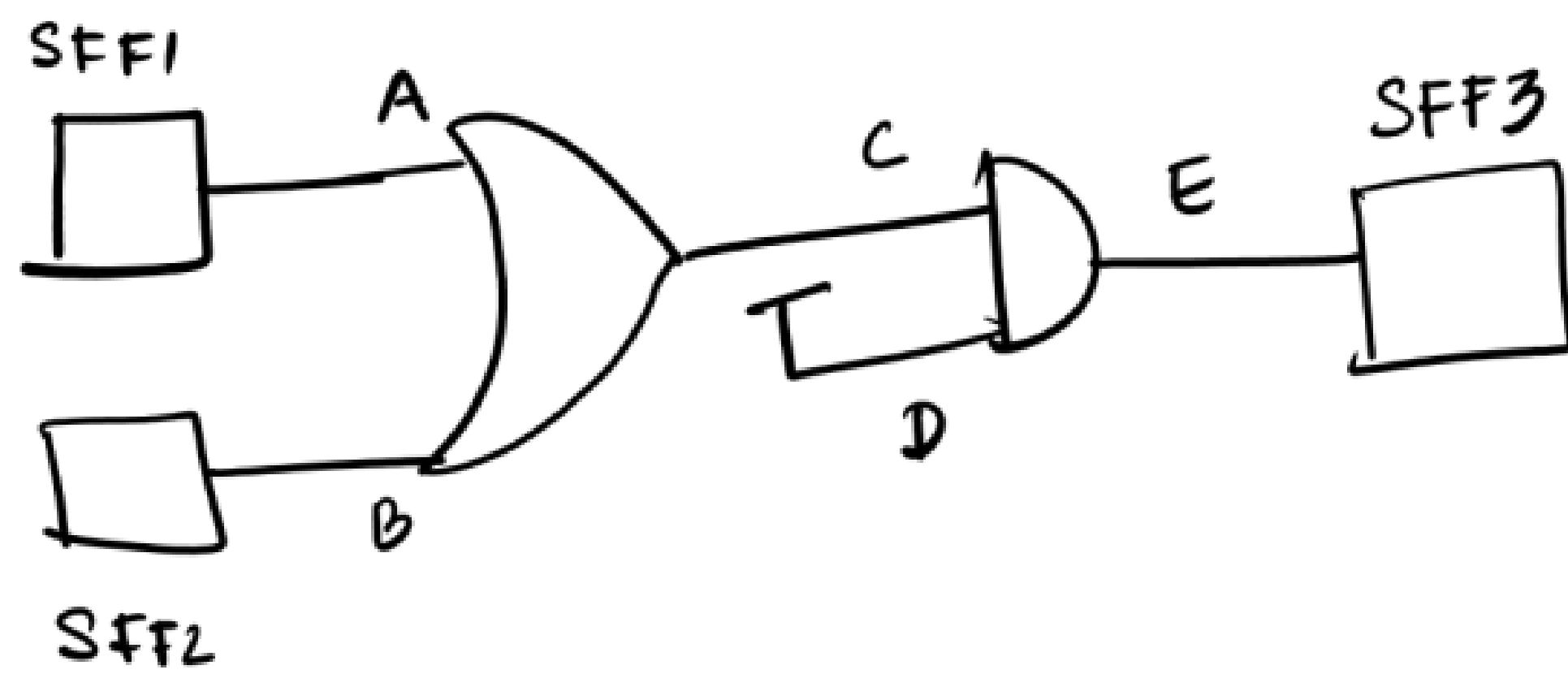
identify opp value: $D=1$

Propagate the value at the fault site to an obs pt

$C=1$
 $\uparrow$
 $A = 0 \text{ or } 1 \text{ or } 1$
 $B = 1 \text{ or } 0 \text{ or } 1$

$Gm/c \neq Fm/c$





DSA1:

Assume a fault: DSA1

identify opp value: $D = 0$

But in this case, tool will not be able to bring 0 at D because D is tied to VDD in the design itself.

$\therefore$ DSA1 will not be detected

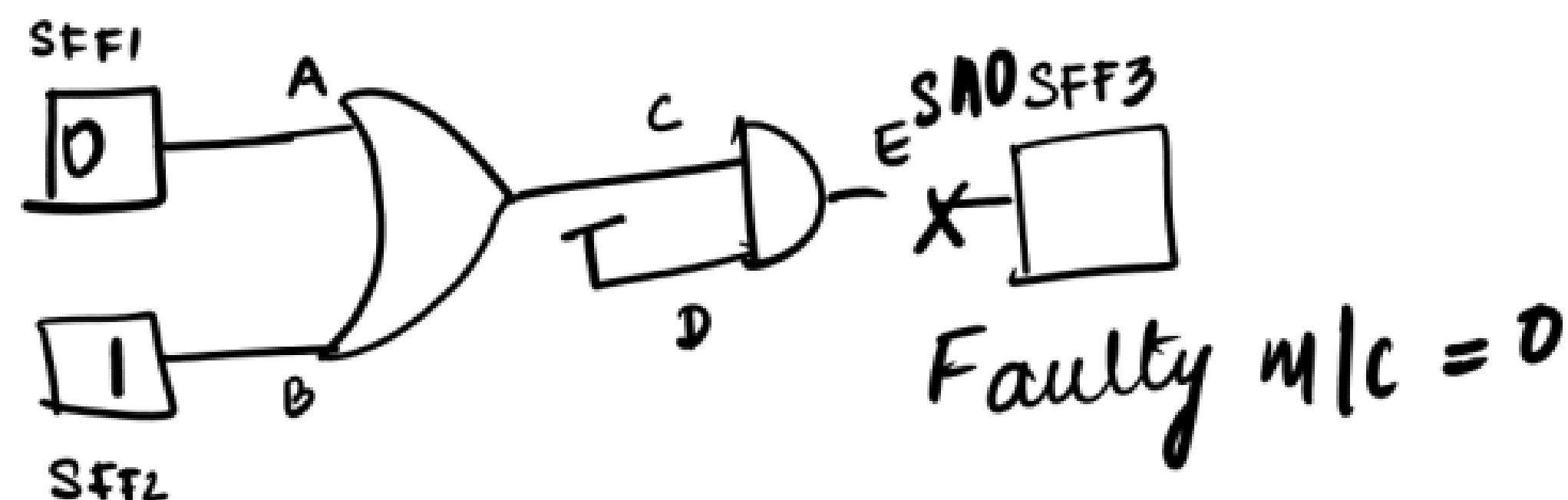
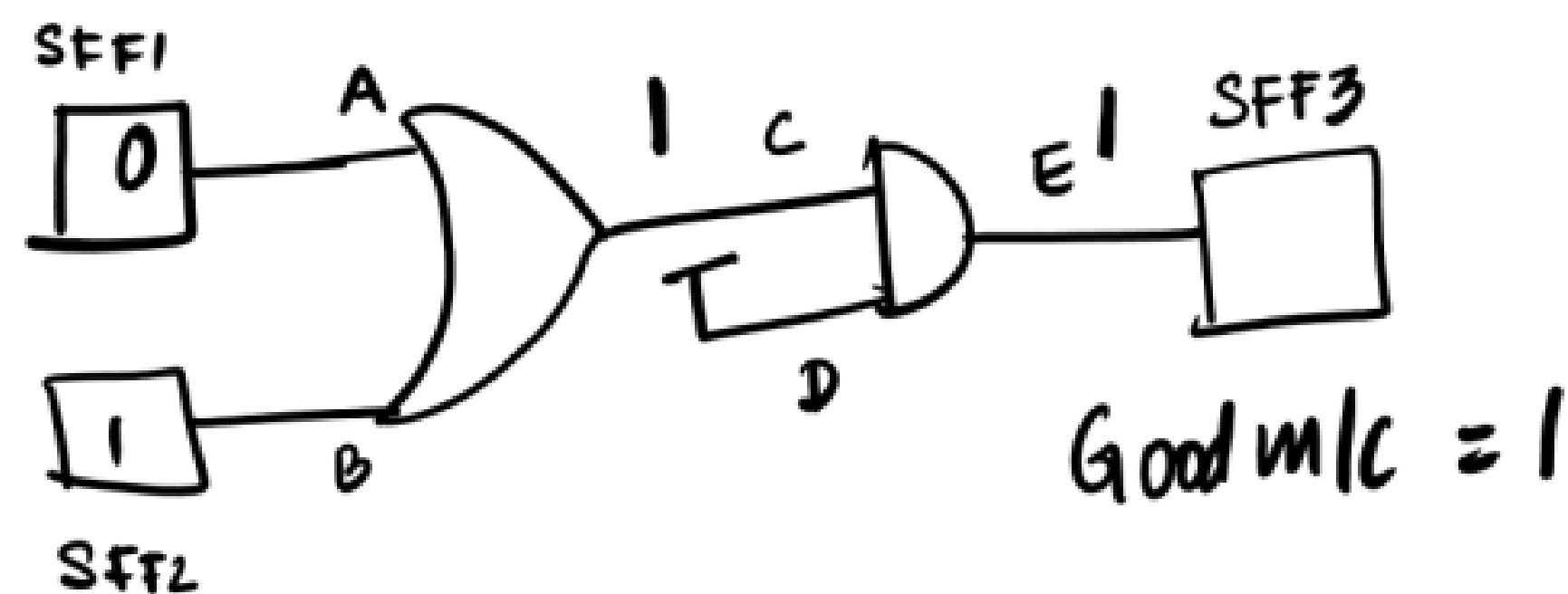
ESAO:

Assume a fault: ESAO

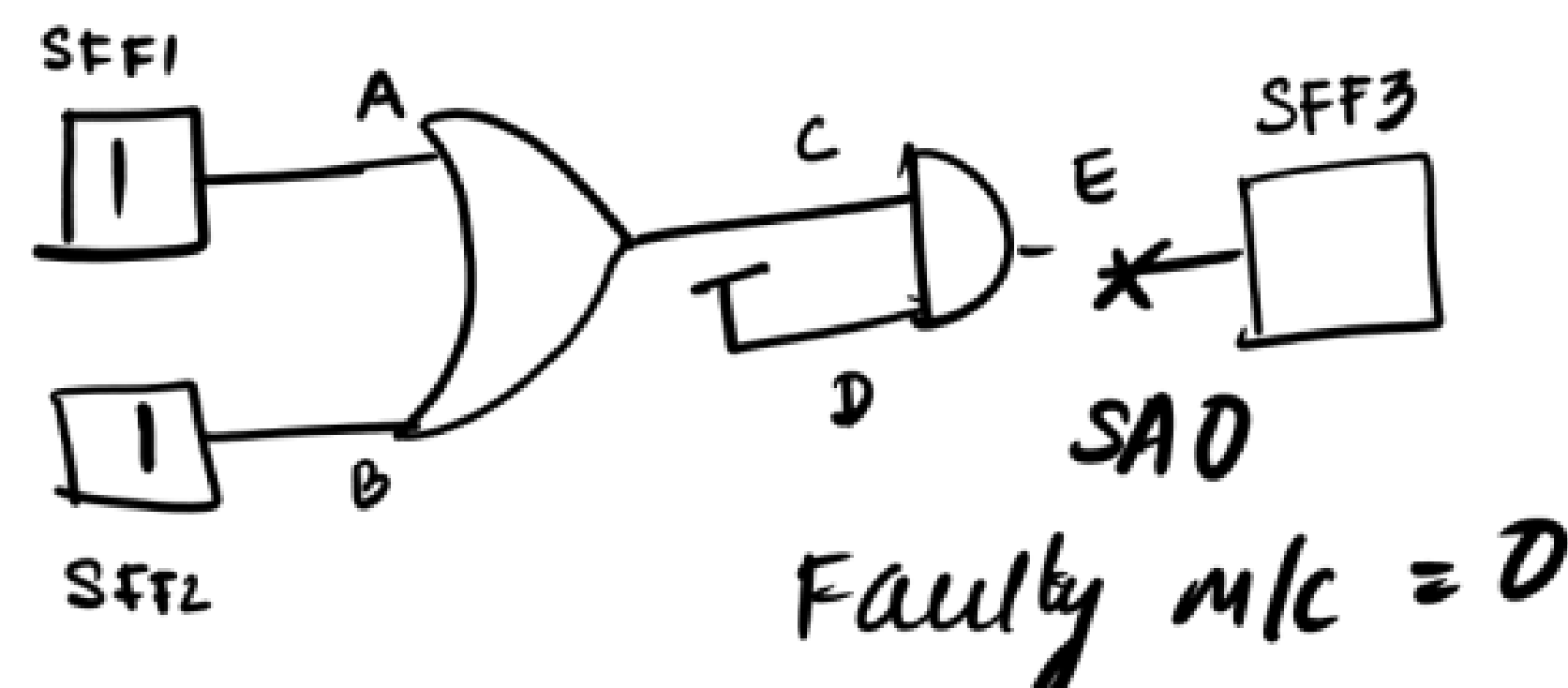
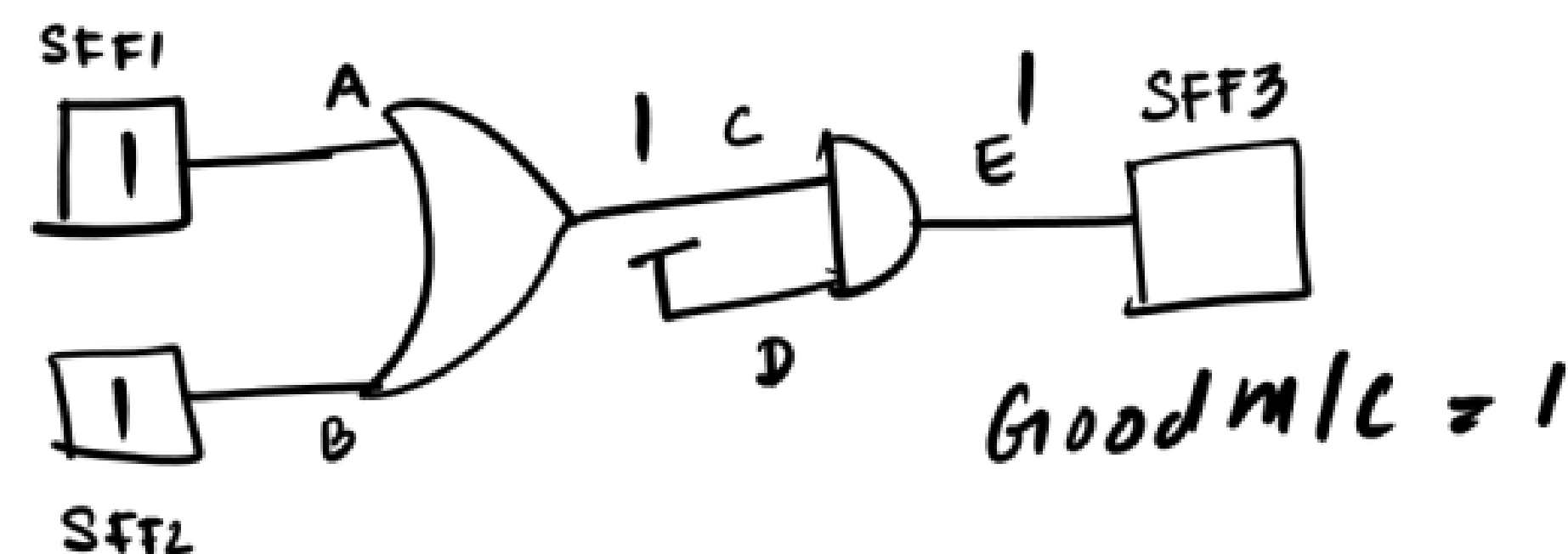
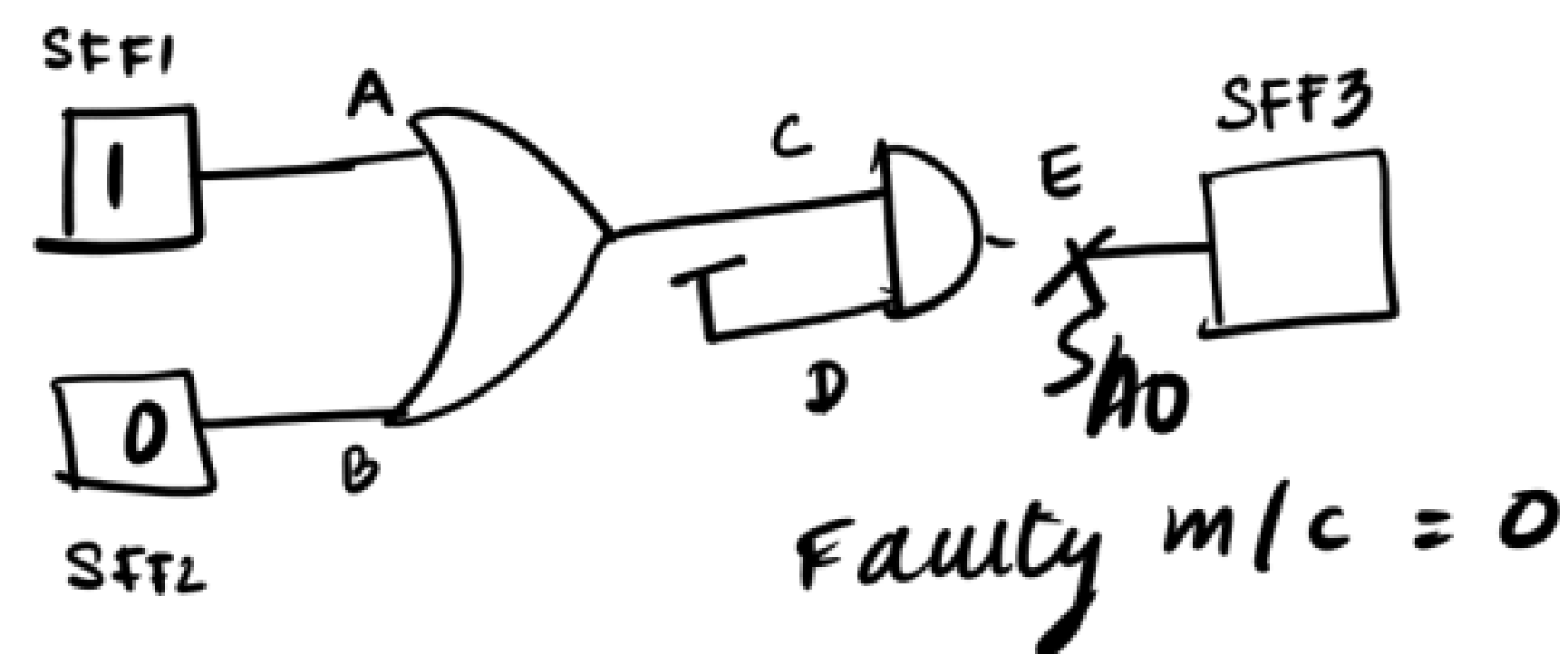
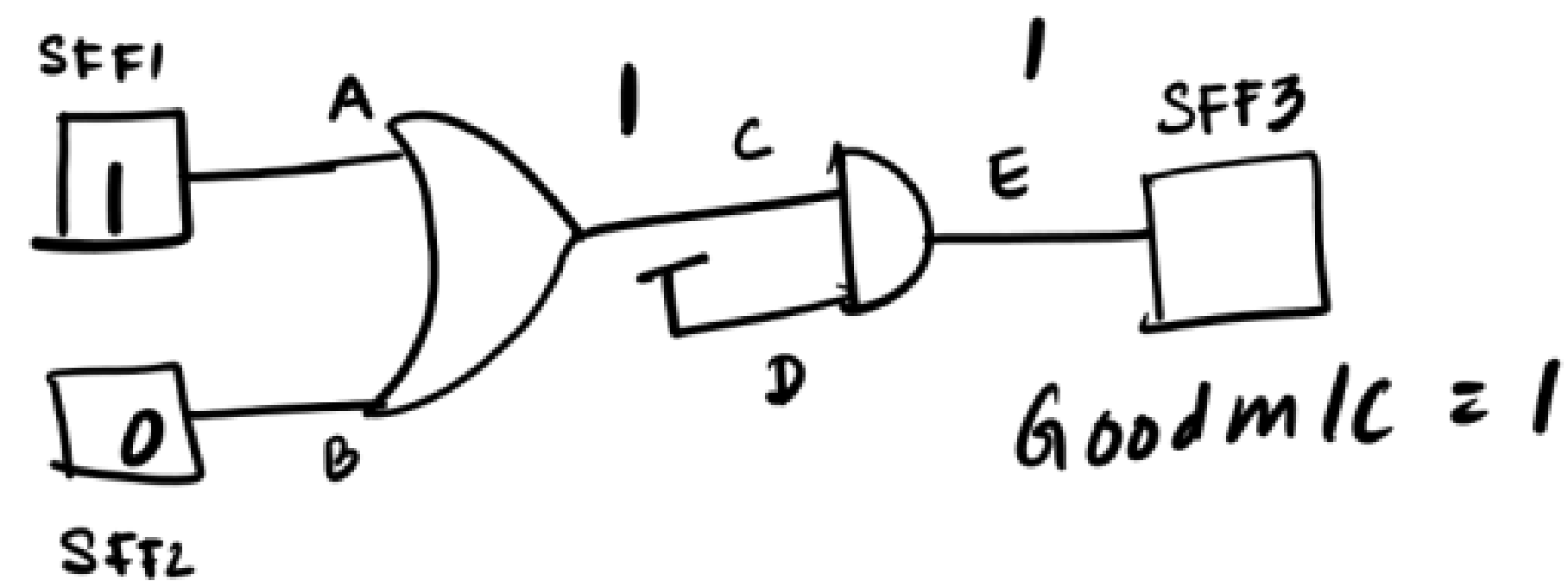
Identify opp value: $E = 1$

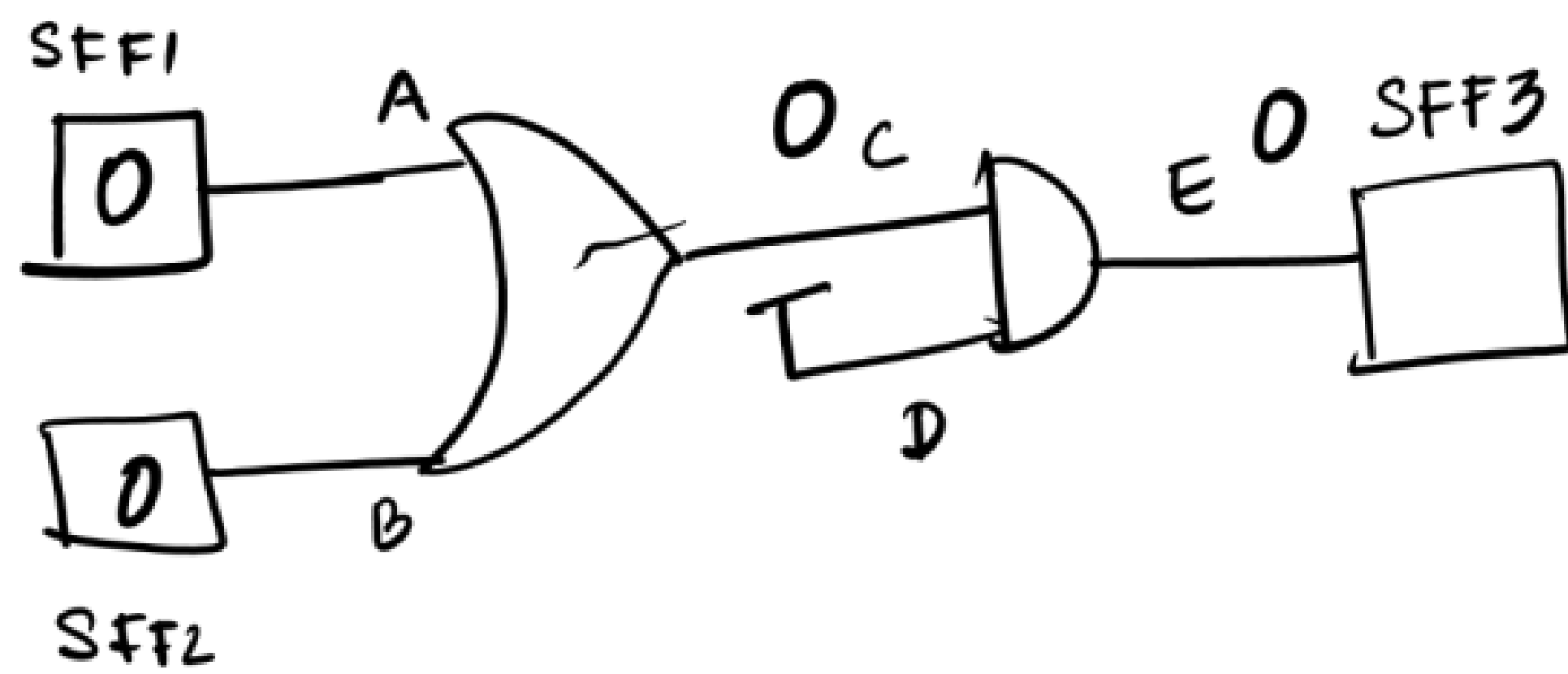
Write i/p seq to get the opp value

$C = 1 \rightarrow A = 0$ or 1
 $D = 1$ $B = 1$ or 0

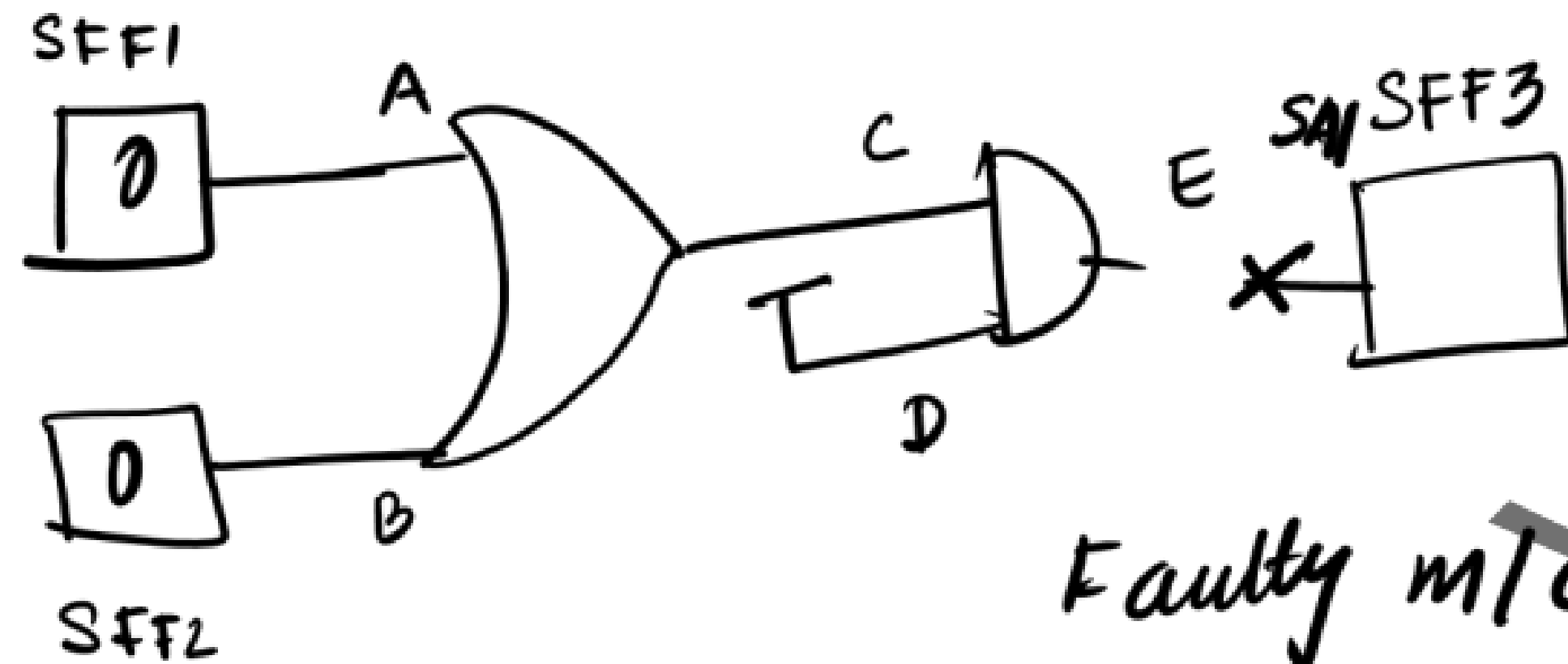


$Gm/c \neq Fm/c$





$$Gm/c = 0$$



$$\text{Faulty } m/c = 1$$

$$Gm/c \neq Fm/c$$

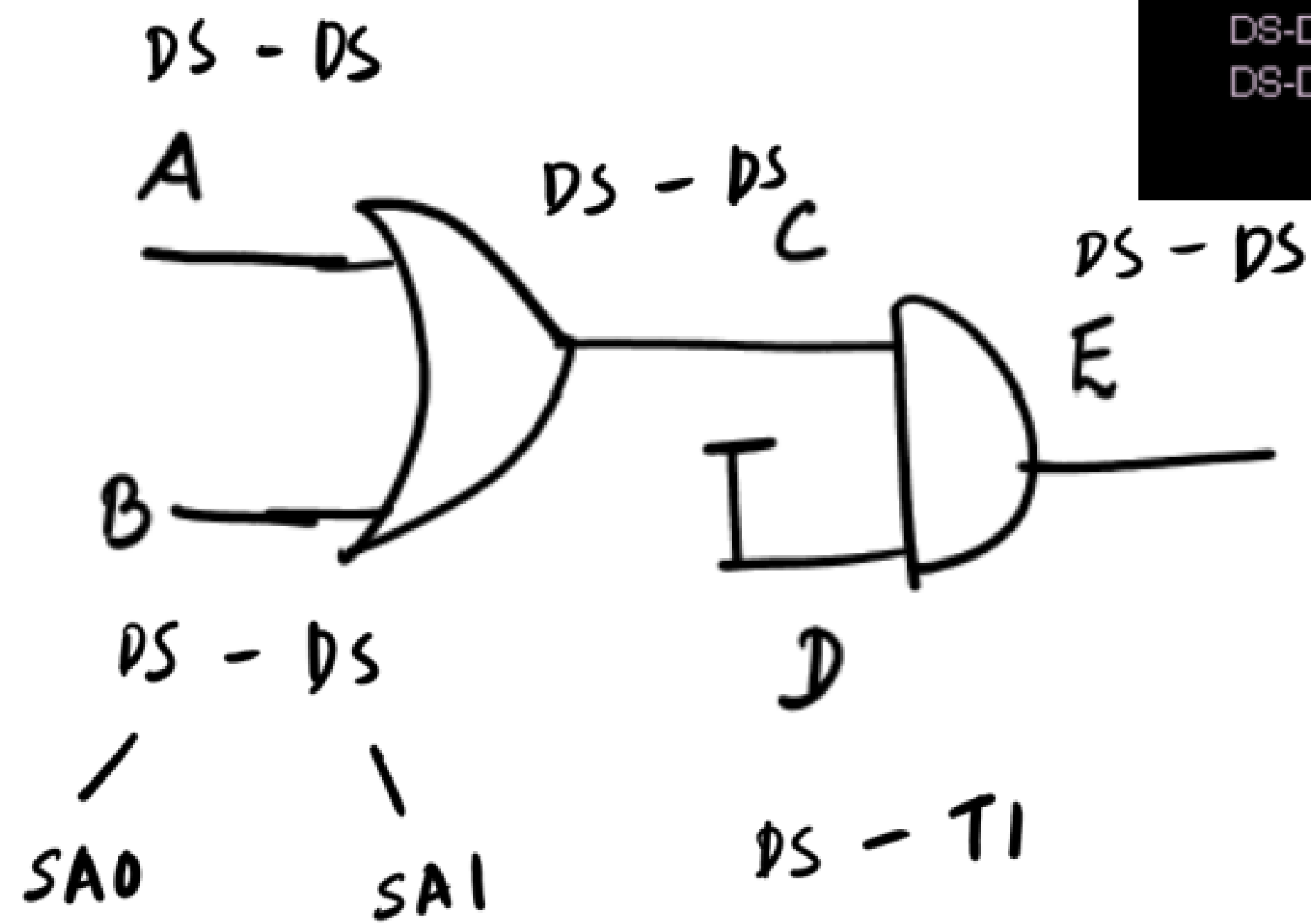
Assume a fault: ESA1
 identify opp value: $E = 0$

write the i/p seq to get the
 opp value

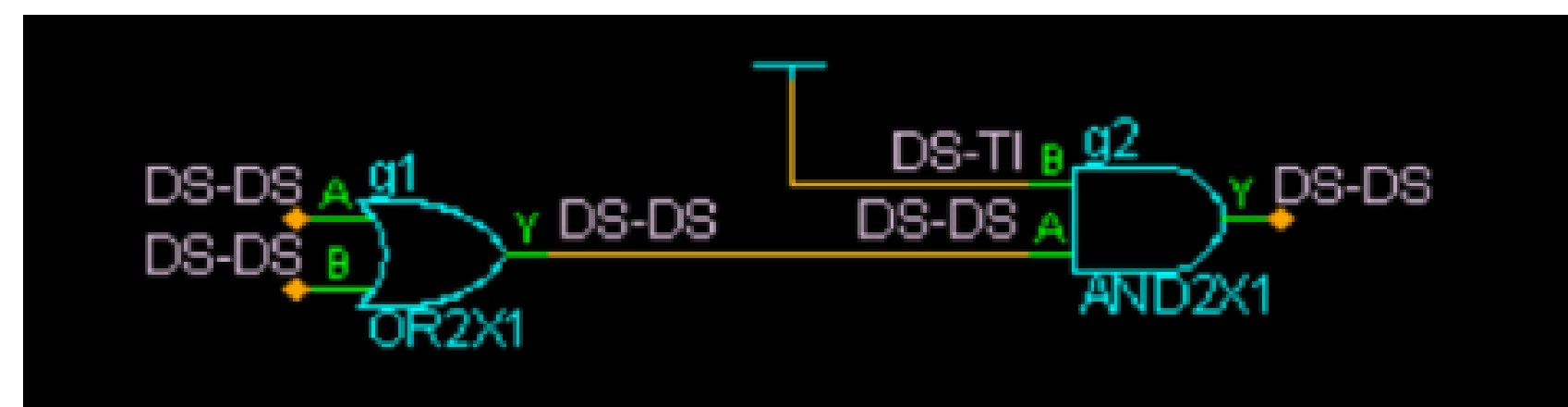
$D = 1$ (since it is tied to vdd
 in the design itself)

$$C = 0 \rightarrow A = 0$$

$$B = 0$$



$$\text{Test cov} = \frac{9}{9} \times 100 = 100\%$$



FAULT

TESTABLE

UNTESTABLE

- Det
- Pos Det
- ATPG Untestable
- Undetected

- UU
- TI
- BL
- RE

$$\text{Fault cov} = \frac{9}{10} \times 100 = 90\%$$